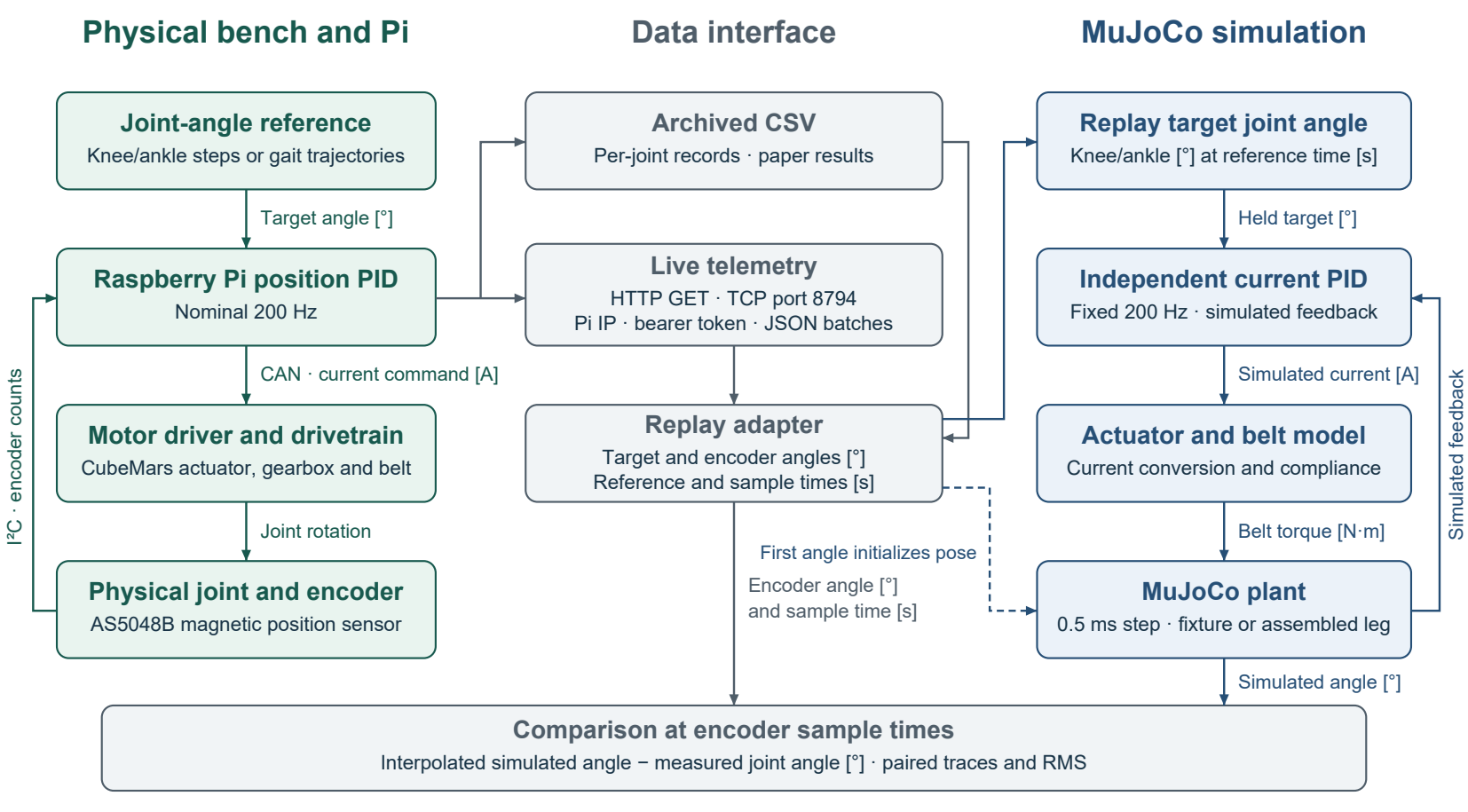

Physical bench and Pi
Data interface
MuJoCo simulation
Joint-angle reference
Knee/ankle steps or gait trajectories
Target angle [°]
Raspberry Pi position PID
Nominal 200 Hz
CAN · current command [A]
Motor driver and drivetrain
CubeMars actuator, gearbox and belt
Joint rotation
Physical joint and encoder
AS5048B magnetic position sensor
I²C · encoder counts
Archived CSV
Per-joint records · paper results
Live telemetry
HTTP GET · TCP port 8794
Pi IP · bearer token · JSON batches
Replay adapter
Target and encoder angles [°]
Reference and sample times [s]
First angle initializes pose
Encoder angle [°]
and sample time [s]
Replay target joint angle
Knee/ankle [°] at reference time [s]
Held target [°]
Independent current PID
Fixed 200 Hz · simulated feedback
Simulated current [A]
Actuator and belt model
Current conversion and compliance
Belt torque [N·m]
MuJoCo plant
0.5 ms step · fixture or assembled leg
Simulated angle [°]
Simulated feedback
Comparison at encoder sample times
Interpolated simulated angle − measured joint angle [°] · paired traces and RMS